

ACCIDENT DETECTION REPORT

DEEP LEARNING PROJECT

PROJECT REPORT

BY

ANEENA VP

BESANT TECHNOLOGIES



**Besant
Technologies**

CONTENTS

INTRODUCTION
ABSTRACT
OBJECTIVES
METHODOLOGY
DATASET OVERVIEW
DATA ANALYSIS & EDA VISUALS
MODEL DEVELOPMENT & RESULTS
STREAMLIT WEB APP DEPLOYMENT
CONCLUSION & FUTURE SCOPE

1. INTRODUCTION

Road accidents are a major concern worldwide and often result in injuries, property damage, and loss of life. Rapid identification of accidents is important to ensure quick emergency response and effective traffic management. With the widespread use of CCTV cameras in cities and highways, automated accident detection systems have become an important application of computer vision.

Traditionally, accident detection relies on manual monitoring or delayed reports from witnesses, which may increase response time. Advances in Deep Learning and Computer Vision allow machines to automatically analyse images and detect unusual traffic situations with high accuracy.

This project focuses on developing a Convolutional Neural Network (CNN) based deep learning model to detect road accidents from CCTV images. The trained model can classify images into accident and non-accident categories and enables automated accident detection for improved road safety.

2. ABSTRACT

This project focuses on detecting road accidents from CCTV images using a Convolutional Neural Network (CNN) based deep learning model. Accident detection plays an important role in improving road safety, traffic monitoring, and emergency response systems. Early identification of accidents can help reduce response time and prevent further damage.

The objective of this project is to build an image classification model capable of identifying accident and non-accident scenarios from traffic images. The system performs image preprocessing, model training, and prediction using deep learning techniques. The trained model learns important visual patterns such as vehicle collisions, road conditions, and abnormal traffic situations.

Additionally, a simple user interface was developed to allow users to upload images and obtain predictions in real time. This project demonstrates how deep learning can be used for intelligent traffic monitoring and automated accident detection.

3. OBJECTIVES

- Build a CNN-based image classification model for accident detection
- Perform image preprocessing and normalization
- Train the model using accident and non-accident image datasets
- Evaluate the model performance using accuracy
- Visualize model performance during training
- Develop a Streamlit web application for real-time accident detection

4. METHODOLOGY

Step 1: Data Collection

- Dataset sourced from Kaggle - accident and non-accident Dataset
- Images represent various traffic scenarios captured from CCTV cameras
- Two classes included: Accident and Non Accident

Step 2: Data Preprocessing

- Images resized to 128×128 pixels
- Images converted into arrays using OpenCV
- Pixel values normalized between 0 and 1
- Labels assigned for accident and non-accident categories

Step 3: Data Splitting

- Dataset divided into training and testing sets
- 80% of the data used for training
- 20% of the data used for testing

Step 4: Model Building

A Convolutional Neural Network (CNN) model was developed.

The architecture consists of:

- Convolutional Layers for feature extraction
- MaxPooling layers for dimensionality reduction
- Flatten layer to convert features into vectors
- Dense fully connected layer with ReLU activation
- Dropout layer to reduce overfitting
- Output layer using sigmoid activation for binary classification

Step 5: Model Evaluation

The model was evaluated using:

- Training accuracy
- Validation accuracy
- Accuracy graphs during training

Step 6: Deployment

A Streamlit web application was developed to allow users to:

- Upload images
- Get predicted No Accident or Accident Occurrence

5. DATASET OVERVIEW

- **Source:** Kaggle – Accident Detection Image Dataset
- **Image Size:** 128×128 RGB images
- **Total Classes:** 2

Classes:

- Accident
- Non Accident

Target Variable: Accident Status

6. DATA ANALYSIS & EDA VISUALS

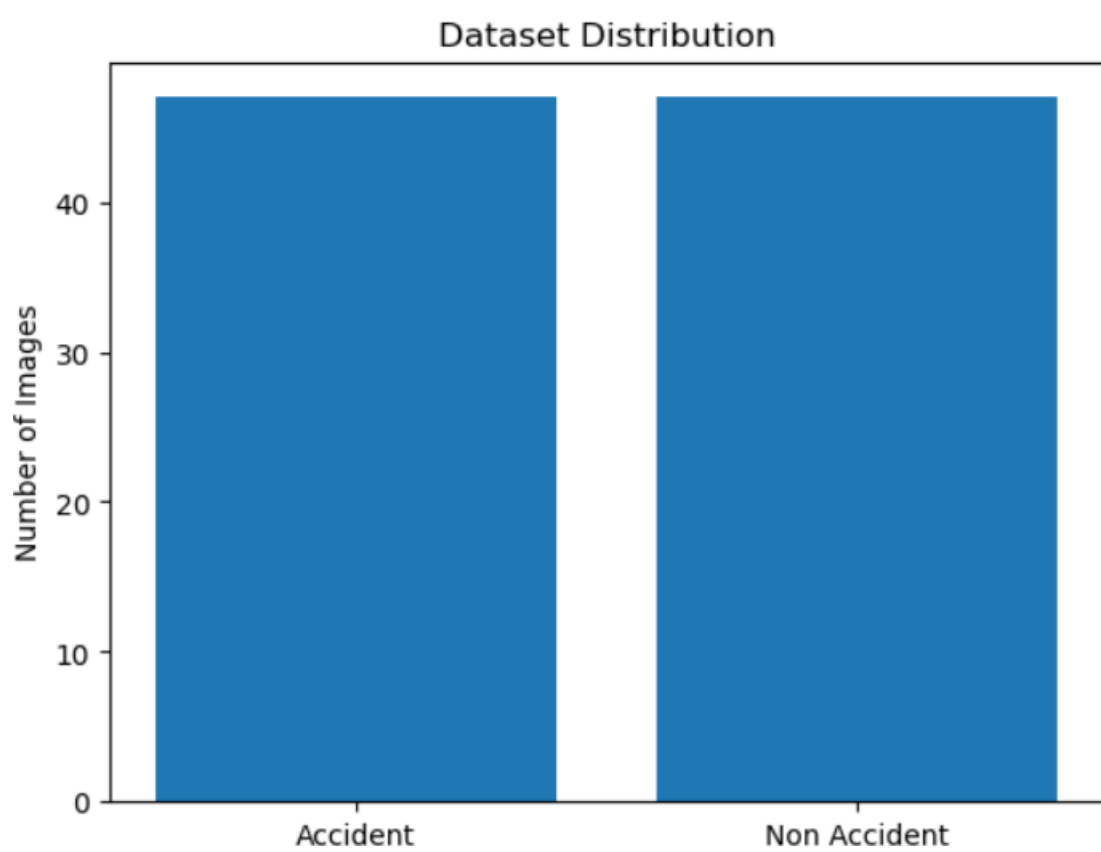
Key observations:

- The dataset contains images of both accident and normal traffic situations
- Accident images contain damaged vehicles, collisions, or unusual road conditions
- Non-accident images represent normal traffic flow and road activity

Visualizations Used:

- Sample image visualization
- Class Distribution Bar Chart





7. MODEL DEVELOPMENT & RESULTS

Model Applied:

Convolutional Neural Network (CNN) using TensorFlow/Keras

- **Model Architecture:**
 - ~Convolutional layers for feature extraction
 - ~MaxPooling layers for reducing image dimensions
 - ~Flatten layer for feature conversion
 - ~Dense layer with ReLU activation
 - ~Dropout layer for regularization
 - ~Output layer using sigmoid activation
- **Compilation:**
 - ~Optimizer → Adam
 - ~Loss Function → Binary Crossentropy
 - ~Evaluation Metric → Accuracy

- **Training Configuration:**

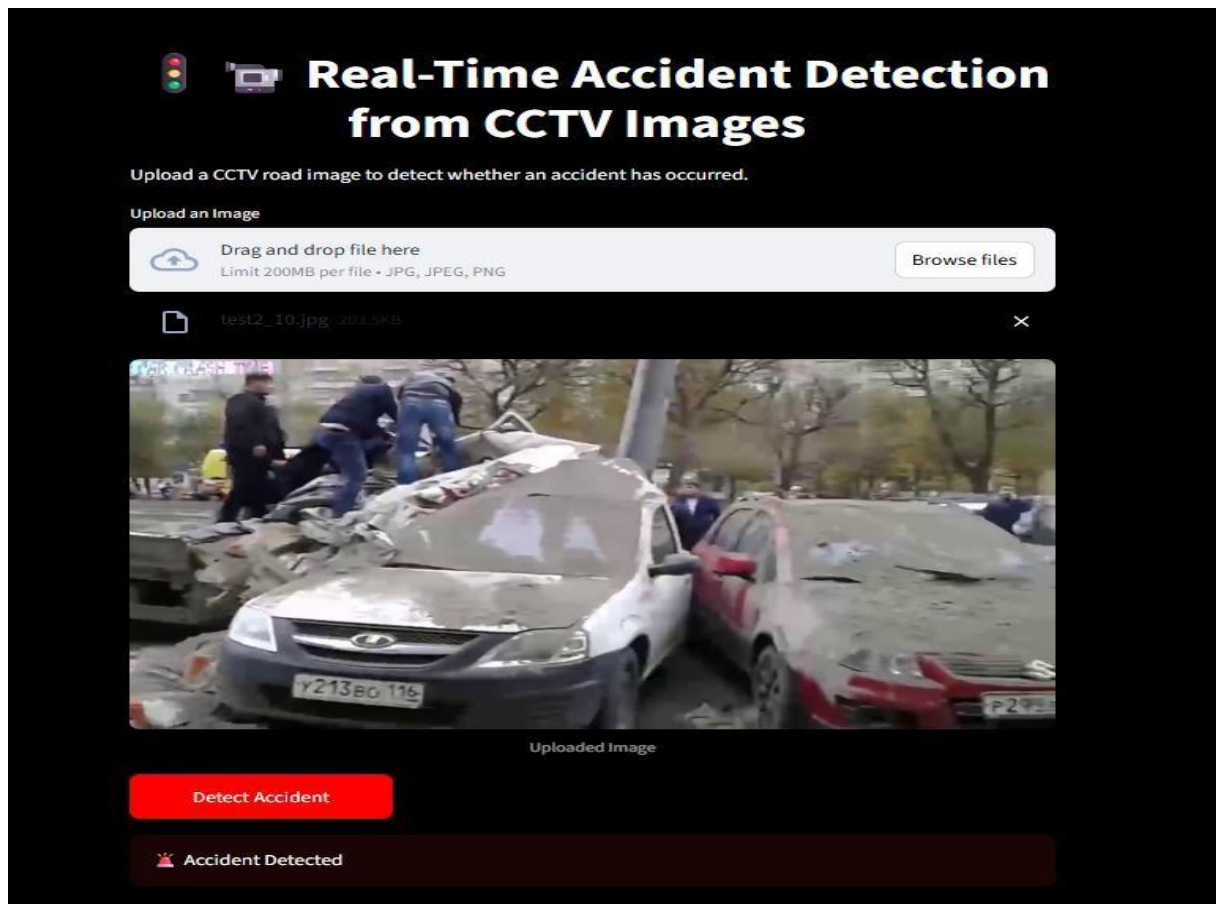
~Epochs → 30

~Batch Size → 3

Performance:

- The CNN model was able to learn visual features from accident and non-accident images effectively.
- Training accuracy improved steadily during model training.
- The validation results indicate that the model performs well in identifying accident scenarios from traffic images.

8. STREAMLIT WEB APP DEPLOYMENT



9. CONCLUSION & FUTURE SCOPE

Conclusion:

The CNN-based deep learning model successfully detects accident scenarios from traffic images. The model can classify images into accident and non-accident categories with good performance. This project demonstrates the potential of deep learning in improving traffic monitoring and road safety systems.

Future Scope:

- Use larger and more diverse datasets
- Implement real-time accident detection from video streams
- Apply advanced deep learning models such as ResNet or EfficientNet
- Integrate the system with smart traffic monitoring infrastructure
- Deploy the model using cloud platforms for large-scale applications